

Definition

Let A be an $n \times n$ matrix. A *quadratic form* defined by A is the function

$$q_A: \mathbb{R}^n \rightarrow \mathbb{R}$$

given by $q_A(\mathbf{v}) = \mathbf{v}^T A \mathbf{v}$.

Example.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$q_A: \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$q_A\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \cdot \begin{bmatrix} x_1 + 2x_2 \\ 3x_1 + 4x_2 \end{bmatrix}$$

$$= x_1(x_1 + 2x_2) + x_2(3x_1 + 4x_2)$$

$$= x_1^2 + 2x_1x_2 + 3x_1x_2 + 4x_2^2$$

$$= x_1^2 + 5x_1x_2 + 4x_2^2$$

Proposition

Let A be an $n \times n$ matrix, and let $A_S = \frac{1}{2}(A + A^T)$. Then:

- 1) A_S is a symmetric matrix.
- 2) $q_A(v) = q_{A_S}(v)$ for all $v \in \mathbb{R}^n$.

Proof:

$$1) \quad A_S^T = \left(\frac{1}{2}(A + A^T) \right)^T = \frac{1}{2} (A + A^T)^T = \frac{1}{2} (A^T + A) = A_S$$

$$\begin{aligned} 2) \quad q_{A_S}(v) &= v^T A_S v = v^T \left(\frac{1}{2} A + \frac{1}{2} A^T \right) v \\ &= \frac{1}{2} v^T A v + \frac{1}{2} v^T A^T v \\ &= \frac{1}{2} v^T A v + \frac{1}{2} (v^T A^T v)^T \\ &= \frac{1}{2} v^T A v + \frac{1}{2} v^T A v \\ &= v^T A v = q_A(v) \end{aligned}$$

Example.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad q_A \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) = x_1^2 + 5x_1x_2 + x_2^2$$

$$A_S = \frac{1}{2} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 5/2 \\ 5/2 & 4 \end{bmatrix}$$

$$\begin{aligned} q_{A_S} \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) &= [x_1 \ x_2] \begin{bmatrix} 1 & 5/2 \\ 5/2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = [x_1 \ x_2] \cdot \begin{bmatrix} x_1 + 5/2 x_2 \\ 5/2 x_1 + 4x_2 \end{bmatrix} \\ &= x_1(x_1 + 5/2 x_2) + x_2(5/2 x_1 + 4x_2) \\ &= x_1^2 + \frac{5}{2} x_1 x_2 + \frac{5}{2} x_1 x_2 + 4x_2^2 = x_1^2 + 5x_1 x_2 + 4x_2^2 \\ &= q_A \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) \end{aligned}$$

Upshot. When defining a quadratic form q_A we can always assume that the matrix A is symmetric.

Change of variables in a quadratic form

Recall: If A is an $n \times n$ symmetric matrix then

$$A = QDQ^T$$

where:

$$Q = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \dots & \mathbf{u}_n \end{bmatrix} \quad \text{orthogonal matrix, } Q^T Q = I_n$$

$$D = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix} \quad \begin{array}{l} \lambda_1 = \text{eigenvalue corresponding to } \mathbf{u}_1 \\ \lambda_2 = \text{eigenvalue corresponding to } \mathbf{u}_2 \\ \dots \dots \dots \dots \dots \dots \dots \dots \\ \lambda_n = \text{eigenvalue corresponding to } \mathbf{u}_n \end{array}$$

Upshot. For any vector $\mathbf{v} \in \mathbb{R}^n$ we have

$$q_A(\mathbf{v}) = q_D(Q^T \mathbf{v})$$

$$q_D(\mathbf{v}) = q_A(Q\mathbf{v})$$

$$q_A(\mathbf{v}) = \mathbf{v}^T A \mathbf{v} \quad q_D(Q^T \mathbf{v}) = (Q^T \mathbf{v})^T D (Q^T \mathbf{v}) = \underbrace{\mathbf{v}^T Q D Q^T}_{A} \mathbf{v} = \mathbf{v}^T A \mathbf{v}$$

Note:

$$q_D \left(\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \right) = [x_1 \dots x_n] \cdot \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = [x_1 \dots x_n] \begin{bmatrix} \lambda_1 x_1 \\ \vdots \\ \lambda_n x_n \end{bmatrix} \\ = \lambda_1 x_1^2 + \lambda_2 x_2^2 + \dots + \lambda_n x_n^2$$

Application: Classification of quadratic forms

Definition

Let A be an $n \times n$ matrix. The quadratic form q_A is

- *positive definite* if $q_A(\mathbf{v}) > 0$ for all $\mathbf{v} \in \mathbb{R}^n - \{0\}$
- *positive semidefinite* if $q_A(\mathbf{v}) \geq 0$ for all $\mathbf{v} \in \mathbb{R}^n - \{0\}$
- *negative definite* if $q_A(\mathbf{v}) < 0$ for all $\mathbf{v} \in \mathbb{R}^n - \{0\}$
- *negative semidefinite* if $q_A(\mathbf{v}) \leq 0$ for all $\mathbf{v} \in \mathbb{R}^n - \{0\}$
- *indefinite* if q_A has both positive and negative values.

Lemma

If D is a diagonal matrix

$$\begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

Then q_D is:

- positive definite if $\lambda_i > 0$ for $i = 1, \dots, n$
- positive semidefinite if $\lambda_i \geq 0$ for $i = 1, \dots, n$
- negative definite if $\lambda_i < 0$ for $i = 1, \dots, n$
- negative semidefinite if $\lambda_i \leq 0$ for $i = 1, \dots, n$
- indefinite if $\lambda_i > 0$ and $\lambda_j < 0$ for some i, j .

Proof: This is clear since

$$q_D \left(\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \right) = \lambda_1 x_1^2 + \lambda_2 x_2^2 + \dots + \lambda_n x_n^2$$

Lemma

Let A be a symmetric matrix with an orthogonal diagonalization

$$A = QDQ^T$$

If the quadratic form q_D is positive definite (positive semidefinite etc.) then q_A has the same property.

Proof: (For positive semidefinite, other cases are similar)

If $q_D(v) \geq 0$ for all $v \in \mathbb{R}^n \setminus \{0\}$

then $q_A(v) = q_D(Q^T v) \geq 0$ for all $v \in \mathbb{R}^n \setminus \{0\}$.

Proposition

Let A be a symmetric matrix with eigenvalues $\lambda_1, \dots, \lambda_n$ then the quadratic form q_A is

- positive definite if $\lambda_i > 0$ for $i = 1, \dots, n$
- positive semidefinite if $\lambda_i \geq 0$ for $i = 1, \dots, n$
- negative definite if $\lambda_i < 0$ for $i = 1, \dots, n$
- negative semidefinite if $\lambda_i \leq 0$ for $i = 1, \dots, n$
- indefinite if $\lambda_i > 0$ and $\lambda_j < 0$ for some i, j .

Example. Classify the quadratic form q_A defined by the matrix

$$A = \begin{bmatrix} 1 & 1 & 5 \\ 1 & 5 & 1 \\ 5 & 1 & 1 \end{bmatrix}$$

Solution:

Characteristic polynomial of A

$$\begin{aligned} P(\lambda) &= \det \begin{bmatrix} 1-\lambda & 1 & 5 \\ 1 & 5-\lambda & 1 \\ 5 & 1 & 1-\lambda \end{bmatrix} = -\lambda^3 + 7\lambda^2 + 16\lambda + 112 \\ &= (7-\lambda)(4-\lambda)(-4-\lambda) \end{aligned}$$

$$(\text{Eigenvalues of } A) = (\text{roots of } P(\lambda)) = \{\lambda_1 = 7, \lambda_2 = 4, \lambda_3 = -4\}$$

Thus q_A is indefinite.

Proposition

Let A be a symmetric matrix. The quadratic form q_A is positive semidefinite if and only if there exists a matrix B such that $A = B^T B$.

Proof:

We have seen before: if $A = B^T B$ then all eigenvalues of A are ≥ 0 , so q_A is positive semidefinite

Conversely, if q_A - positive semidefinite then

$$A = Q D Q^T, \quad D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}, \quad \lambda_i \geq 0$$

$$\text{Define } \sqrt{D} = \begin{bmatrix} \sqrt{\lambda_1} & & 0 \\ & \ddots & \\ 0 & & \sqrt{\lambda_n} \end{bmatrix} \text{ and let } B = Q \sqrt{D} Q^T$$

We have:

$$B^T B = (Q \sqrt{D} Q^T)^T (Q \sqrt{D} Q^T)$$

$$= Q \sqrt{D}^T Q^T \cdot Q \sqrt{D} Q^T$$

$$\begin{aligned} &= Q \sqrt{D}^T \sqrt{D} Q^T &= Q \sqrt{D} \cdot \sqrt{D} \cdot Q^T \\ &\quad \uparrow &\quad \uparrow \\ &Q^T Q = I &\sqrt{D}^T = \sqrt{D} \\ & &= Q D Q^T = A \end{aligned}$$

Constrained optimization of quadratic forms

Constrained Maximum Problem. Given a symmetric matrix A , find a vector $\mathbf{v} \in \mathbb{R}^n$ such that $\|\mathbf{v}\| = 1$ and the value $q_A(\mathbf{v})$ is the largest possible.

Lemma

If D is a diagonal matrix

$$D = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

such that $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$, then the vector $\mathbf{e}_1 = \begin{bmatrix} 1 & 0 & \dots & 0 \end{bmatrix}^T$ is a solution of the Constrained Maximum Problem. Also, $q_D(\mathbf{e}_1) = \lambda_1$

Proposition

Let A be a symmetric matrix with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$. If $\mathbf{u}_1$ is an eigenvector corresponding to λ_1 such that $\|\mathbf{u}_1\| = 1$ then $\mathbf{u}_1$ is a solution of the Constrained Maximum Problem and $q_A(\mathbf{u}_1) = \lambda_1$.

Example.

$$A = \begin{bmatrix} 1 & 1 & 5 \\ 1 & 5 & 1 \\ 5 & 1 & 1 \end{bmatrix}$$